

Crop Yield Estimation

MILESTONE 2

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PRESENTED BY

Angelica Moreno
Arunima Sen
Ethel Ogallo

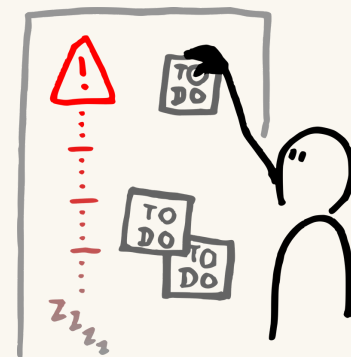
RECAP

Data Overview



Dataset

Synthetically generated dataset simulating real-world agricultural scenarios



Purpose

Created for practicing machine learning models, specifically for **predicting crop yield**

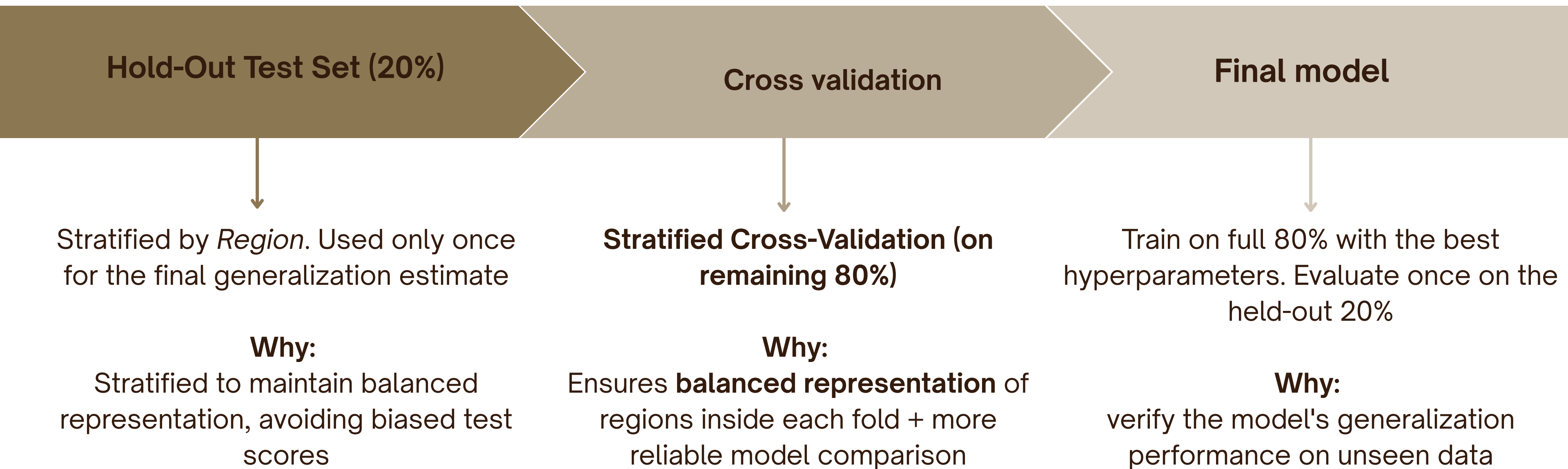


Learning task

Supervised regression

RECAP

Revised evaluation protocol



ML Algorithms

1 - Dummy regressor

Baseline model that predicts a constant value (e.g., the mean).

3 - Histogram-based gradient boosting

Builds trees sequentially where each tree corrects errors from the previous one using histogram-based splits.

2 - Ridge regression

Linear regression with L2 regularization to penalize large coefficients

4 - Random Forest

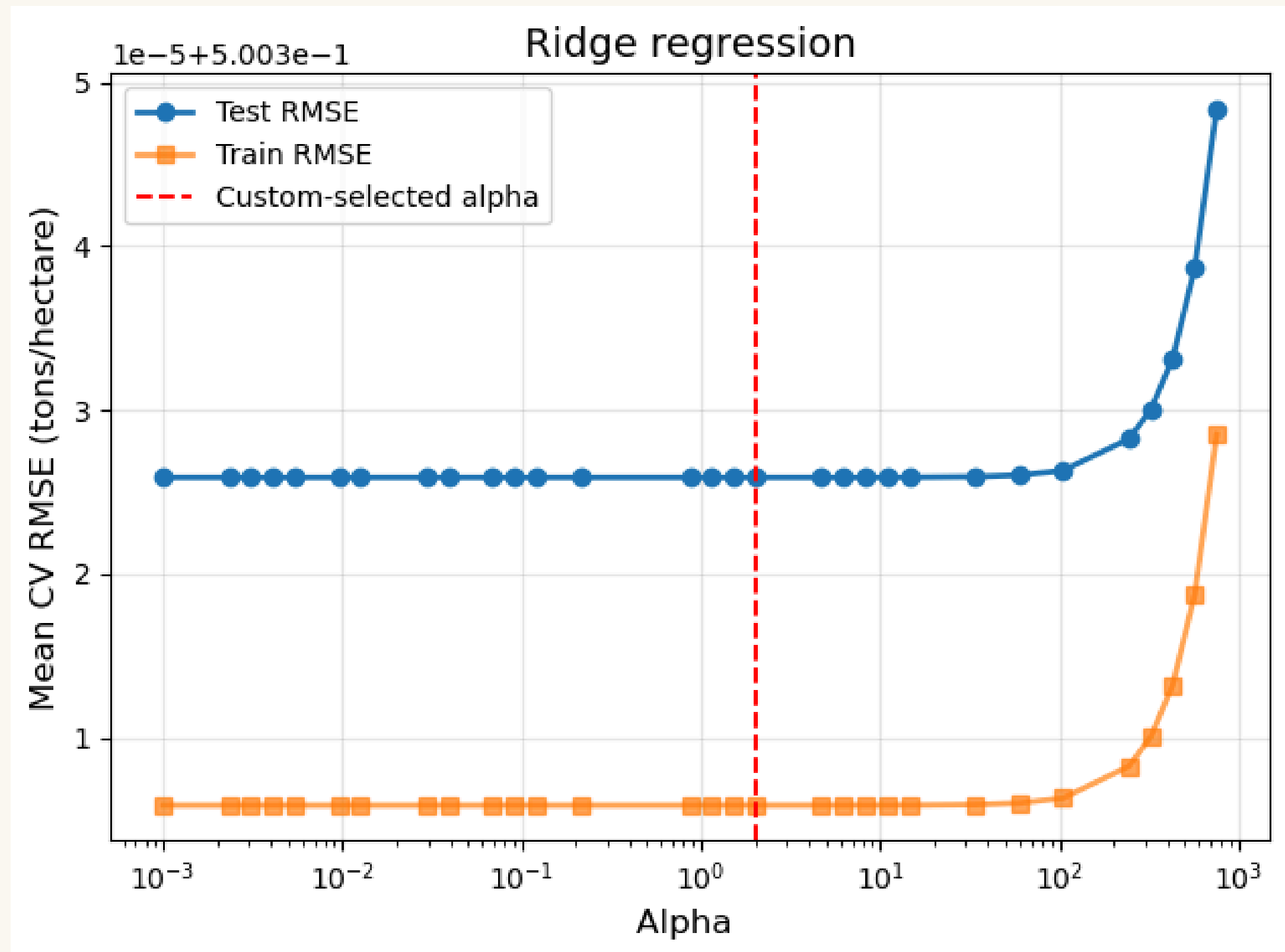
Builds many decision trees on random data/features and averages predictions to reduce variance.

Ridge regression

Best parameter value:

$\alpha \approx 2$

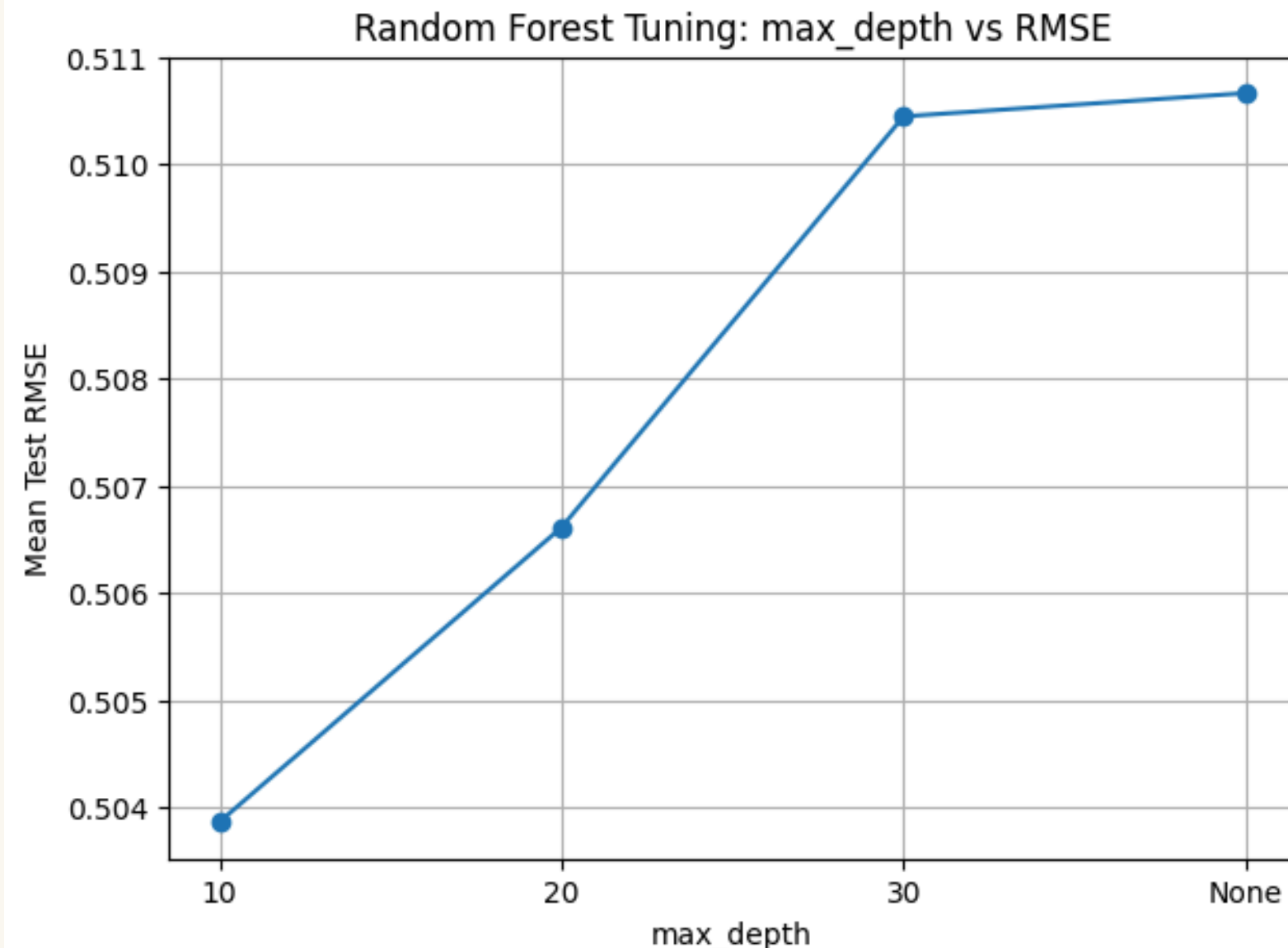
It balances low test RMSE with a minimal train-test gap, providing stable coefficients and reducing the risk of overfitting compared to the very small initial alpha.



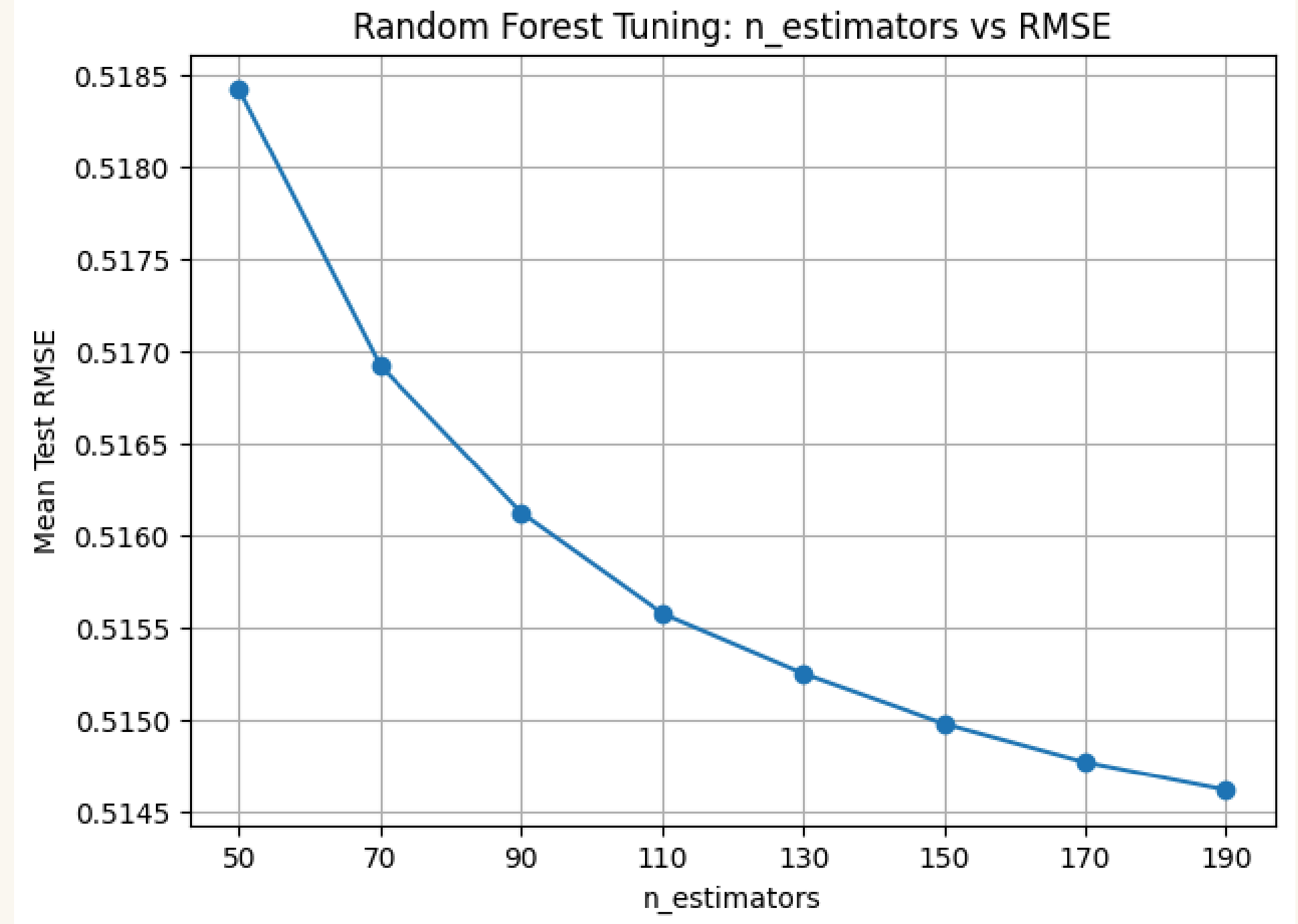
Random Forest

Best Parameter values:

- `n_estimators=190`,
- `max_features=0.5`,
- `max_depth=10`,
- `min_samples_leaf=4`,
- `max_samples=0.1`,



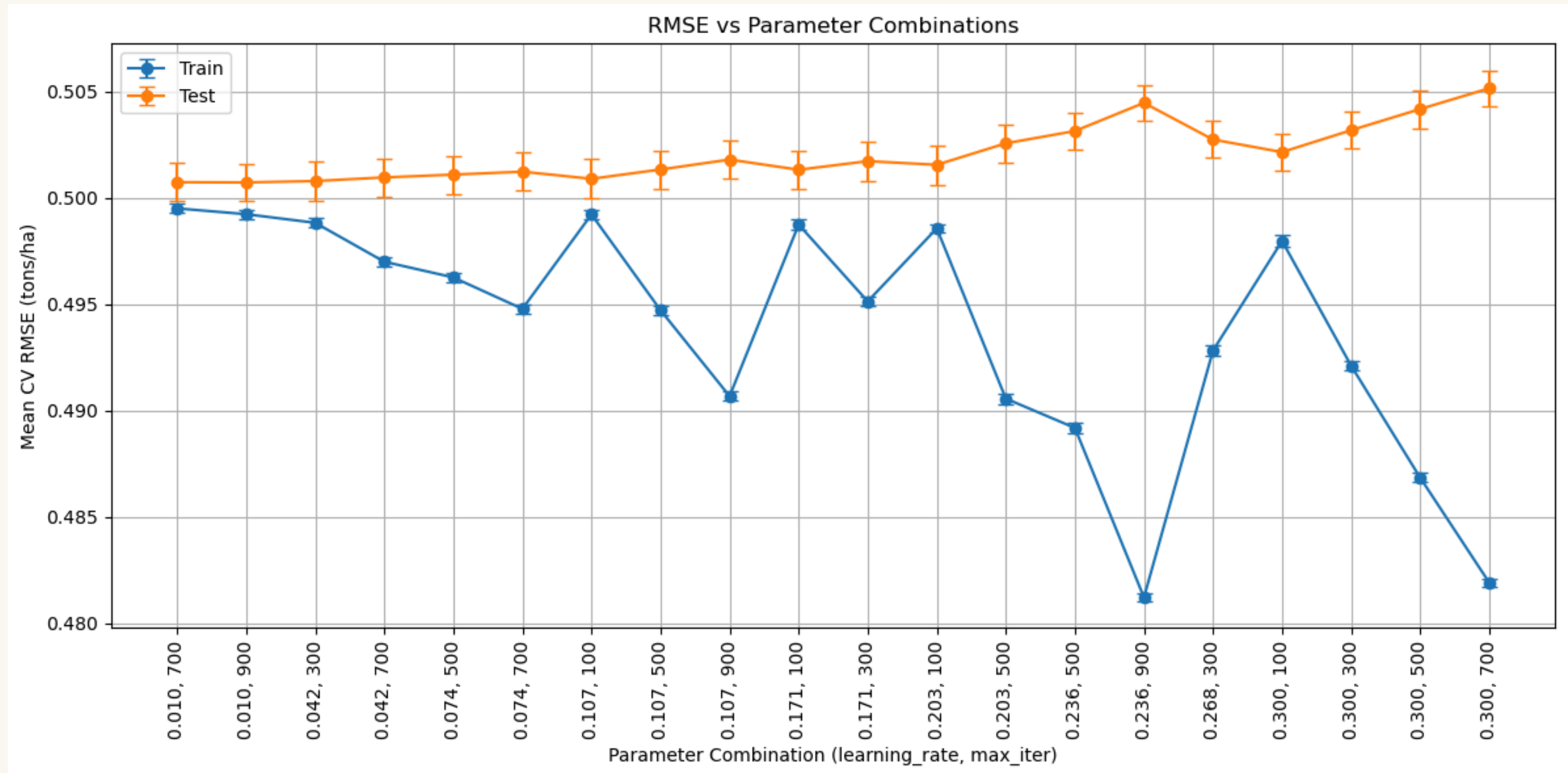
HYPERPARAMETER TUNING



Histogram-based gradient boosting

Best parameters:

- learning_rate = 0.01
- max_iter = 700
- max_leaf_nodes = 15
- Early stopping = False

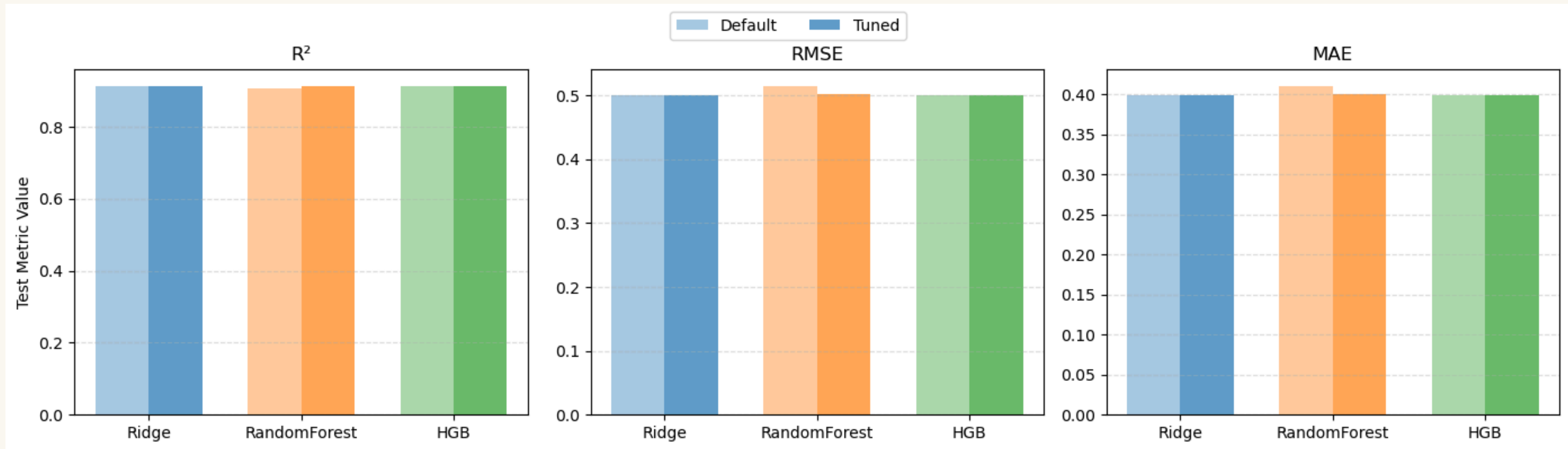


Model Comparison

MODEL COMPARISON

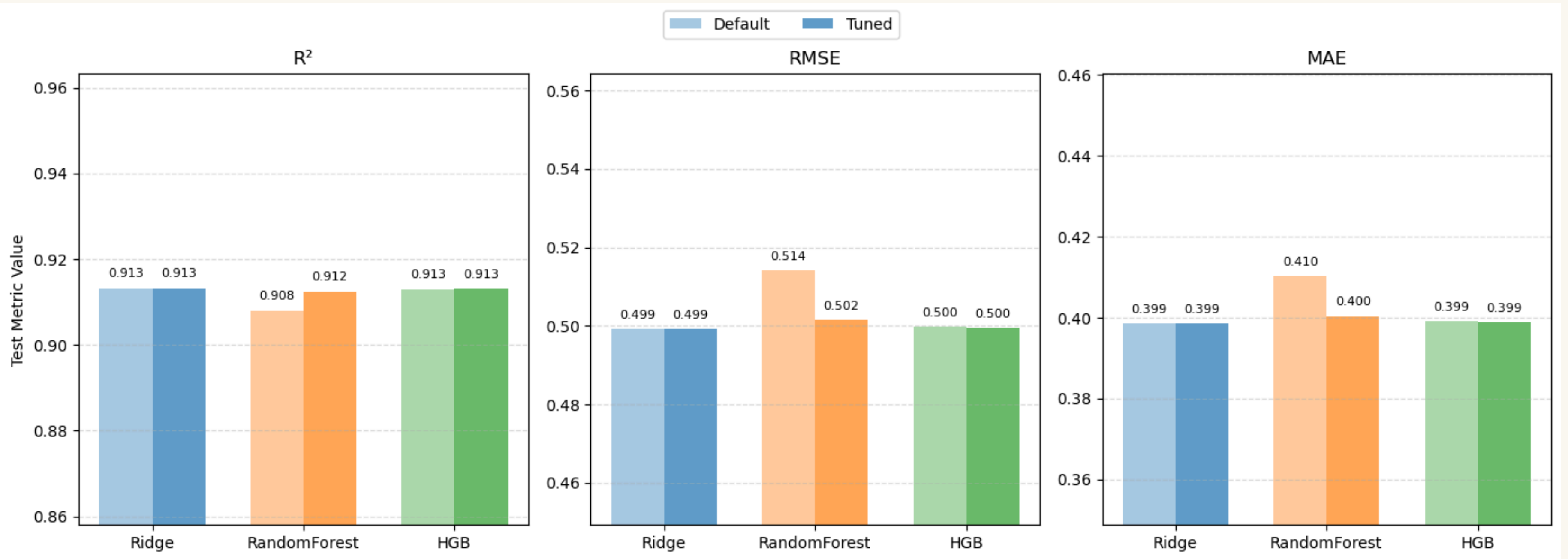
Default vs Tuned model results:

Difference in the test scores of the models with the default parameters and tuned parameters on the three models is not as significant.



MODEL COMPARISON

Closer look



MODEL COMPARISON

ML Algorithm	Advantages	Disadvantages
Dummy Regressor	Provides a baseline and easy to implement.	Cannot capture any patterns in the data e.g $R^2 \approx 0$, RMSE ≈ 1.6949
Ridge Regressor	handles multicollinearity and fast to train	Limited to linear relationships
Histogram-based Gradient Boosting	Can capture non-linear relationships and efficient because of histogram-based splits	Can overfit if learning rate too high
Random Forest Regressor	Captures complex non-linear patterns and robust after tuning	Longer training time and can overfit

Model choice

Ridge Regression

Why:

- It achieves the lowest RMSE score of **0.4993 tonnes/ha**
- It is also computationally the most efficient.

Limitation

Data is synthetic and does not reflect real-world variability

- Add realistic noise to simulate environmental and measurement variability to test model generalization